Can Desktop Virtual Reality Enhance a Story-Driven Experience?

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COLLEGE OF THE Holy Cross

MOTIVATION

Understand the importance of narratives and graphics in a virtual story-driven experience, and their relevance in advertising, marketing, computer gaming, and filmmaking.

RESEARCH QUESTION

How do **narrative quality** and **graphics quality** affect users in a **story-driven experience**?

INTRODUCTION

Literature Review

- A goal of media creators who use **narratives** is to maximize engagement through high **presence** and **enjoyment** (Green et al., 2004)
- Higher **graphics quality** can positively affect a user's experience when they are not otherwise engaged (Newman et al., 2022)
- People are capable of becoming invested in stories with extremely limited visual assistance (Heider & Simmel, 1944)

Research Hypotheses

1. Higher narrative quality will result in more enjoyment, attitude change, and narrative transportation than lower narrative quality.
2. Higher graphics quality will result in more enjoyment, attitude change, and narrative transportation than lower graphics quality.
3. Effects of graphics quality will be stronger for lower than higher narrative quality.

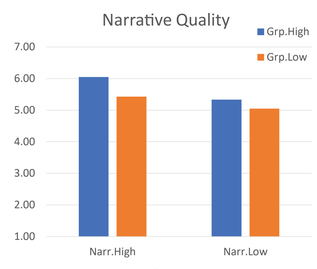
IVs

Narrative Quality	Graphics Quality: High	Graphics Quality: Low
High	High Narrative High Graphics	High Narrative Low Graphics
Low	Low Narrative High Graphics	Low Narrative Low Graphics

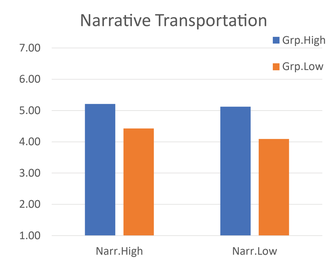
METHOD

Participants 12 undergraduate students
VR System Desktop with 1080p 32" monitor, GTX 1080 Ti
Procedure Practice Session → Main Session → Surveys
Virtual Environment

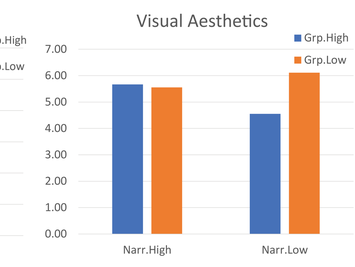
Perception of IV-Related Variables



"I enjoy the fantasy or story provided by the game."

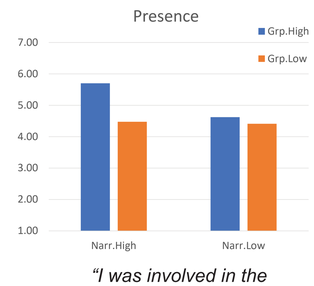


"I could picture myself in the scene of the events described in the narrative."

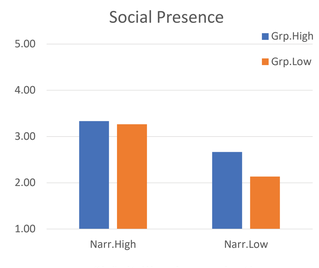


"I think the game is visually appealing."

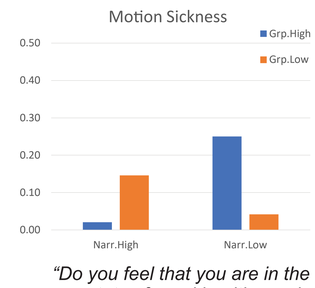
Virtual Environment-Induced Effects



"I was involved in the virtual environment experience completely."



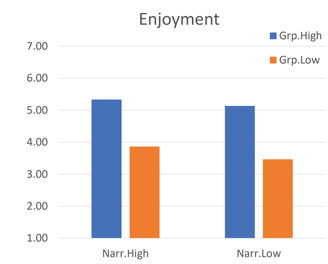
"I felt like I was in the presence of another person in the virtual environment."



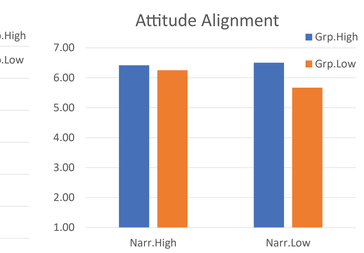
"Do you feel that you are in the same state of good health as when you started the experiment?"

RESULTS

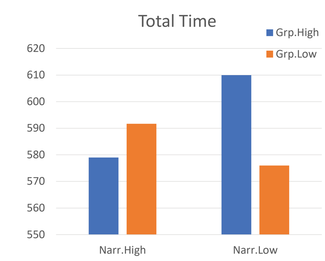
Effect on Cognition and Behavior

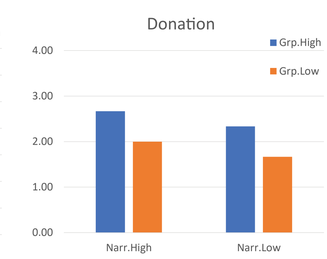


"I enjoy playing the game."



"No one else perceives the world exactly like you do."





Overall story comprehension accuracy: 86.67%

0 = Less than $20
1 = Between $20 and $100
2 = Between $100 and $1,000
3 = Between $1,000 and $10,000
4 = Greater than $10,000

DISCUSSION

Pilot data suggests:

1. Interaction between narrative quality and graphics quality for **total time taken**, **presence**, **social presence**, **attitude alignment**, **narrative quality**, **visual aesthetics**, and **motion sickness**.
2. Increasing graphics quality enhances story-driven experiences related to **enjoyment**, **donation**, and **narrative transportation**.
3. Both narrative quality and graphics quality affect story-driven experiences but graphics quality has stronger effects.